

Multiple Choice (10 marks)

1. Supposing that the clonal selection theory is correct, how can an organism genetically make such diverse proteins?
 - (a) Through vertical gene transfer from parent to offspring
 - (b) Through random assortment of genes during meiosis
 - (c) Through peptide bond formation errors during translation
 - (d) Through genetic mutation only
 - (e) Through transcriptional errors in RNA processing
2. Convergent evolution is best exemplified by which of the following?
 - (a) The gills of a fish and the lungs of a human
 - (b) The leaves of an oak tree and the spines of a cactus
 - (c) The eyes of a human and the eyes of a squid
 - (d) The roots of a tree and the legs of an animal
 - (e) The presence of a notochord in all chordate embryos
3. Which of the following pieces of evidence most strongly supports the common origin of all life on Earth?
 - (a) All organisms reproduce.
 - (b) All organisms are capable of evolving.
 - (c) All organisms require energy.
 - (d) All organisms have DNA.
 - (e) All organisms are composed of cells.
4. What function does oxygen serve in the body?
 - (a) Oxygen is used to maintain the temperature of cells.
 - (b) Oxygen is used by cells to produce alcohol.
 - (c) Oxygen is used to produce carbon dioxide in cells.
 - (d) Oxygen is solely responsible for stimulating brain activity.
 - (e) Oxygen is used to synthesize proteins in cells.
5. What are some of the cues used by animals in migrating?
 - (a) Animals use the temperature changes as cues.
 - (b) Animals use the color of the surroundings as cues.

- (c) Animals use the noises from the environment as cues.
- (d) Animals navigate by following other migrating animal species exclusively.
- (e) Animals rely solely on the memory of previous migration paths.

True / False (10 marks)

Answer True or False

- 6. True or False: The progeny of an O type and an AB type individual will all have AB blood type.
- 7. True or False: Guttation is explained by the effect of root pressure, which forces water out of specialized structures in leaves.
- 8. True or False: Enzymes typically require ionizing groups to maintain their activity across various pH levels.
- 9. True or False: The probability of drawing the ace of diamonds from a standard deck of cards is $1/52$.
- 10. True or False: The pH of both lakes will decrease in response to acid rain, indicating increased acidity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss how you would experimentally demonstrate that a newly discovered bioluminescent worm, rather than contaminating bacteria, is responsible for the light production.
- 12. Discuss the physiological adaptations of whales that enable them to thrive in cold aquatic environments, focusing on thermoregulation and insulation mechanisms.

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